

TradeGuard AI — v5.0 LIVE TRADING INTELLIGENCE LAYER

“Your AI Council, Live in the Room With You”

ADDENDUM TO v4.0 — Add This Entire Module to the Existing Platform

THE CORE IDEA

When a retail trader has an open position, they are alone. They have a chart, a prayer, and maybe a Telegram group full of bad advice. This module changes that permanently.

Module 8: The Live Trading Copilot is a full-screen intelligence panel that activates the moment a trader enters a position. Six AI models simultaneously run parallel research from six completely different perspectives — technical, fundamental, institutional, social, psychological, and mathematical — and deliver a unified live briefing in real time.

It is the closest thing that has ever existed to having a Bloomberg Terminal analyst, a trading psychologist, a quant, and a dark pool researcher all sitting next to you watching the same trade.

NEW DATA INTEGRATIONS (Add to Existing Stack)

Dark Pool & Options Flow (The Institutional Layer)

- **Unusual Whales API** — real-time options flow, dark pool prints, congressional trades
- **FlowAlgo API** (or Quant Data API) — dark pool block prints, unusual option sweeps
 - Dark pool prints = off-exchange block trades that signal institutional positioning
 - Option sweeps = aggressive multi-exchange options buys that signal urgency
 - Block trades = large institutional equity trades outside normal order books

On-Chain Intelligence (Crypto Layer)

- **Glassnode API** — BTC/ETH on-chain metrics (exchange in-flows/outflows, whale movements)
- **Nansen API** — wallet labeling, smart money tracking, whale alerts

Technical Data

- **Alpaca WebSocket** — tick-by-tick real-time price streaming
- **OANDA Streaming API** — forex tick data
- **Binance WebSocket** — crypto real-time orderbook + trades

Gamma Exposure (Derivatives Layer)

- **SpotGamma API** (or Quant Data) — real-time GEX (Gamma Exposure), DEX, VEX, CHEX
 - GEX = dealer gamma exposure (predicts price pinning or volatility expansion)
 - DEX = delta exposure (net directional bias of options market makers)
 - Vanna/Charm = second-order Greeks driving intraday flows
-

MODULE 8: LIVE TRADING COPILOT

8.1 — Architecture: The Parallel Research Engine

When a position is opened (detected via broker webhook or manual entry), the system immediately fires **6 parallel async AI research jobs** — one per perspective. Results stream back to the frontend via Socket.io as each completes.

Position Opened → Trigger Event →

Job A: Technical Analysis	→ GPT-o3 (math-heavy)
Job B: Institutional Flow	→ Claude Opus 4 (synthesis)
Job C: Dark Pool Intelligence	→ Claude Opus 4 + FlowAlgo data
Job D: Social Sentiment	→ Grok 4 (real-time X)
Job E: Fundamental Context	→ Perplexity (cited) + Gemini (deep)
Job F: Behavioral Monitor	→ Claude Opus 4 (psych analysis)

↓

All results → Socket.io → Live Copilot Panel (frontend)

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Every 60 seconds: re-run all 6 jobs with fresh data

↓

On price move >1%: immediate re-run of all jobs

↓

On news event for ticker: immediate re-run of Job D + Job E

8.2 — The 6 Research Perspectives

PERSPECTIVE 1: TECHNICAL INTELLIGENCE Powered by:
GPT-o3 (mathematical precision) Data inputs: Real-time OHLCV, volume, order book snapshot

Calculates ALL of the following in real time and updates every 60 seconds:

PRICE STRUCTURE:

- Current price vs key levels: 20/50/200 EMA, VWAP, daily/weekly/monthly pivots
- Distance to nearest support and resistance (in \$ and %)
- Whether price is above/below VWAP (institutional benchmark)
- Current candlestick pattern on 1m, 5m, 15m, 1h timeframes

MOMENTUM INDICATORS:

- RSI (14) across 4 timeframes - identify divergence
- MACD histogram direction + signal line cross status
- Stochastic RSI - overbought/oversold on current timeframe
- ADX (14) - trend strength (>25 = trending, <20 = ranging)
- Williams %R - confirmation signal

VOLATILITY METRICS:

- ATR (14) - current daily range vs average
- Bollinger Band width - squeeze detection
- Historical volatility (20-day HV) vs implied volatility (IV)
- IV percentile (IVP) - is current IV historically cheap or expensive
- Keltner Channel position

VOLUME ANALYSIS:

- Current volume vs 20-day average volume (% of average)
- VWAP deviation - how far price is from volume-weighted average
- OBV (On-Balance Volume) trend direction
- Volume Profile: identify high-volume nodes (HVN) and low-volume nodes (LVN)
- Cumulative Delta - buying pressure vs selling pressure (from tick data)

STATISTICAL EDGE:

- Z-score of current price move (how statistically unusual is this move)
- Expected move for remainder of session based on current ATR
- Mean reversion probability at current price level (historical backtest)
- Trend continuation probability (based on ADX + momentum alignment)

OUTPUT FORMAT - GPT-o3 responds with:

```
{
  "overallBias": "BULLISH | BEARISH | NEUTRAL",
  "biasStrength": 0-100,
  "keyLevels": {
    "support1": price, "support2": price,
    "resistance1": price, "resistance2": price,
    "vwap": price, "ema20": price, "ema50": price
  }
}
```

```

    },
    "momentum": { "rsi14": value, "macdBias": "BULLISH|BEARISH", "adxStrength": value },
    "volatility": { "atr14": value, "ivPercentile": value, "bbSqueeze": boolean },
    "volume": { "vsAverage": "HIGH|NORMAL|LOW", "vpocLevel": price, "cumulativeDelta": "BUYING|SELLING",
    "statEdge": { "zScore": value, "expectedMoveRemaining": value, "meanReversionProb": value },
    "immediateAlert": "string or null - urgent technical warning",
    "summary": "2-sentence plain English technical read"
  }
}

```

PERSPECTIVE 2: INSTITUTIONAL FLOW INTELLIGENCE

Powered by: Claude Opus 4 synthesis of FlowAlgo + Unusual

Whales data Data inputs: Dark pool prints, options sweeps, block trades
(last 4 hours for this ticker)

DARK POOL DATA ANALYSIS:

- Dark pool prints for this ticker in the last 4 hours
(size, price, direction estimate - bid/ask side)
- Dark pool volume as % of total volume
(>15% = significant institutional positioning)
- Are dark pool prints clustering near a price level?
(indicates hidden support/resistance)
- Divergence: dark pool direction vs public market price action

OPTIONS FLOW ANALYSIS:

- Unusual options activity for this ticker
(sweeps = urgent institutional buys/sells across multiple exchanges)
- Put/Call ratio vs 30-day average
- Net options premium: bullish \$ vs bearish \$
- Unusual open interest changes on specific strikes
- Options sweeps in the last 2 hours: direction and urgency score

GAMMA EXPOSURE:

- Current GEX level - positive GEX = price pinning (market makers stabilize)
negative GEX = volatility acceleration (market makers amplify moves)
- Key gamma flip level (price where GEX transitions positive to negative)
- Vanna exposure - directional pressure from delta hedging flows
- Charm - time decay pressure on dealer hedging today

SMART MONEY SYNTHESIS (Claude analyzes all the above):

Claude Prompt:

"You are a dark pool and institutional flow analyst.

A retail trader currently holds a [LONG/SHORT] position in {{symbol}} entered at \${{entry}}.

Current price: \${{price}}. Time in trade: {{duration}} minutes.

DARK POOL DATA (last 4 hours):

{{darkPoolPrints}} - [array of prints: size, price, side estimate, time]

Dark pool volume % of total: {{dpVolumePct}}%

OPTIONS FLOW (last 4 hours):

{{optionsSweeps}} - [sweeps: strike, expiry, premium, direction, urgency]

Net options premium: Bullish \${{bullPremium}} vs Bearish \${{bearPremium}}

Put/Call ratio: {{pcRatio}} (30-day avg: {{pcRatioAvg}})

GAMMA EXPOSURE:

Current GEX: {{gex}} ({{gexBias}})

Gamma flip level: \${{gammaFlip}}

Vanna bias: {{vannaBias}}

Analyze what the institutional and smart money activity is saying about this ticker.
Is the smart money aligned or opposed to this retail trader's position?

Respond ONLY with valid JSON:

```
{
  'institutionalBias': 'BULLISH|BEARISH|NEUTRAL|MIXED',
  'confidenceScore': 0-100,
  'darkPoolSignal': 'ACCUMULATING|DISTRIBUTING|NEUTRAL|UNCLEAR',
  'optionsFlowSignal': 'AGGRESSIVELY_BULLISH|BULLISH|NEUTRAL|BEARISH|AGGRESSIVELY_BEARISH',
  'gammaEnvironment': 'PINNING|ACCELERATING|NEUTRAL',
  'gammaFlipLevel': price,
  'smartMoneyVsPosition': 'ALIGNED|OPPOSED|NEUTRAL',
  'keyInsight': '2-sentence most important institutional observation',
  'urgentAlert': 'string or null - something requiring immediate attention',
  'riskToPosition': 'LOW|MEDIUM|HIGH|CRITICAL'
}
```

PERSPECTIVE 3: DARK POOL PRINTS LIVE FEED UI Component: Dedicated panel (not AI — raw data feed with AI annotation)

Real-time stream of dark pool prints for the user's open positions:

Each print displays:

- Time (HH:MM:SS)
- Symbol
- Size (shares/contracts)
- Price
- Side estimate (BUY side / SELL side / UNKNOWN)
- % of average daily volume
- Significance badge:

WHALE (>500K shares or >\$10M notional)
LARGE (>100K shares or >\$2M notional)
NOTABLE (>50K shares)
STANDARD

Claude annotates each WHALE or LARGE print in real time:
"This \$12.4M dark pool print on the BID side suggests a large institution may be distributing (selling) stock near this price level. Combined with the current price action, this could indicate near-term resistance."

PERSPECTIVE 4: SOCIAL & NARRATIVE INTELLIGENCE

Powered by: Grok 4 (real-time X access) Updates: Every 5 minutes during position, immediately on viral post detection

Grok Prompt:

"Search X right now for discussion about {{symbol}}.
The trader has been long since \${{entry}} and is currently at \${{currentPnl}} P&L.

Find:

1. What are traders and analysts on X saying about {{symbol}} right now?
2. Is there any breaking news or rumors being discussed?
3. Are any large accounts (verified, 100K+ followers) posting about this?
4. Is there any coordinated pump or panic behavior in the posts?
5. What is the dominant emotion in the posts: fear, greed, uncertainty, euphoria?
6. Has anything changed in the narrative in the last 30 minutes?

Respond ONLY with valid JSON:

```
{
  'xSentiment': 'VERY_BULLISH|BULLISH|NEUTRAL|BEARISH|VERY_BEARISH',
  'sentimentScore': -1.0 to 1.0,
  'dominantEmotion': 'FEAR|GREED|UNCERTAINTY|EUPHORIA|PANIC|CALM',
  'majorAccounts': [{account, followerCount, postSummary, sentiment}],
  'breakingNarrative': 'string or null - anything going viral right now',
  'pumpOrPanicDetected': boolean,
  'retailFOMOLevel': 0-100,
  'contrarianSignal': 'string or null - is everyone bullish = bearish signal?',
  'summary': '2-sentence social intelligence read',
  'urgentAlert': 'string or null'
}
```

PERSPECTIVE 5: FUNDAMENTAL & NEWS INTELLIGENCE

Powered by: Perplexity (verified facts) + Gemini 2.5 Pro (deep context)

Step 1 - Perplexity verifies:

"Search for breaking news about {{symbol}} in the last 2 hours.
Verify each piece of news from the original source.
Return only verified facts with source URLs.
Flag any unverified rumors."

Step 2 - Gemini synthesizes fundamental context:

"Given these verified news items about {{symbol}}:
{{verifiedNews}}

And this company's fundamentals:

Revenue growth: {{revenueGrowth}}%
P/E ratio: {{pe}} vs sector avg {{sectorPe}}
Next earnings date: {{earningsDate}}
Analyst consensus: {{analystConsensus}} | Price target: \${{priceTarget}}
Institutional ownership %: {{instOwnership}}
Short interest: {{shortInterest}}% (float)
Days to cover: {{daysToCover}}

A retail trader is currently [LONG/SHORT] {{symbol}} at \${{entry}}.
Current price: \${{price}}.

Provide deep fundamental context for this position:

1. Does the current price make fundamental sense relative to valuation?
2. Are there any upcoming catalysts (earnings, FDA dates, product launches) that change the
3. What does the analyst community think and why does it matter for a short-term trade?
4. Is the short interest level a risk factor or opportunity (short squeeze potential)?
5. Is this price level near any historically significant fundamental level?

Respond in JSON format with fundamental bias, catalyst risk, and plain-English context."

PERSPECTIVE 6: LIVE BEHAVIORAL MONITOR Powered by:
Claude Opus 4 — the psychological guardian The most important
perspective. Updates every 60 seconds.

This is not just analysis. This is an AI that watches how you are behaving in
real time and fires warnings BEFORE you make a destructive decision.

Behavioral Monitor tracks in real time:

TRADING BEHAVIOR SIGNALS:

- Time in trade vs user's historical average for this asset class
(holding too long compared to your own pattern = red flag)
- Current P&L vs user's historical "panic sell" level
(do you always sell at -2%? You're at -1.8% right now)

- Current P&L vs user's historical "euphoria exit" level
(do you always hold too long when winning? You're at +3.2%)
- Number of times the user has checked this position in the last 10 minutes
(frequent checking = emotional state indicator)
- Time of day vs user's historical worst trading window
(you lose 70% of trades placed after 3 PM - it's 3:15 PM)
- Consecutive losses today (approaching historical revenge trading threshold)

CONTEXT-AWARE PSYCHOLOGY PROMPT - Claude Opus 4:

"You are a trading psychologist monitoring a retail trader in real time.
This trader is currently in an active position.

TRADER'S HISTORICAL PATTERNS (from their journal data):

- Dominant biases: {{dominantBiases}}
- Historical 'panic sell' level: typically exits losing trades at {{panicExitLevel}}%
- Historical 'euphoria hold' pattern: often holds winners 47% past ideal exit
- Best trading window: {{bestHour}}:00-{{bestHour2}}:00
- Worst trading window: {{worstHour}}:00-{{worstHour2}}:00
- Consecutive loss trigger: historically revenge trades after {{consecutiveLosses}} losses
- Recent journal emotion: last 3 entries showed {{recentEmotions}}

CURRENT SITUATION:

- Asset: {{symbol}} | Direction: {{side}} | Entry: \${{entry}}
- Current P&L: {{pnl}}% (\${{pnlDollar}})
- Time in trade: {{duration}} minutes (avg for this trader: {{avgDuration}} min)
- Time of day: {{currentTime}} (this trader's worst window: {{worstWindow}})
- Consecutive losses today: {{consecutiveLossesToday}}
- This is trade #{{tradeNumber}} today

BEHAVIORAL FLAGS DETECTED:

{{detectedFlags}} - array of triggered behavioral warning conditions

Assess this trader's current psychological state and risk of a destructive decision.
What is the highest probability mistake they are about to make based on their own pattern?

Respond ONLY with valid JSON:

```
{
  'psychState': 'CALM_DISCIPLINED|SLIGHTLY_ANXIOUS|EMOTIONALLY_COMPROMISED|HIGH_RISK|TILT',
  'stateScore': 0-100, // 0 = calm, 100 = full emotional hijack
  'likelyNextMistake': 'PANIC_SELL|IGNORE_STOP|AVERAGE_DOWN|REVENGE_TRADE|EUPHORIA_HOLD|NONE',
  'mistakeProbability': 0-100,
  'warningMessage': '2-sentence direct message to the trader about what you are observing',
  'recommendedAction': 'HOLD_PLAN|REVIEW_STOP|CONSIDER_EXIT|EXIT_NOW|CLOSE_PLATFORM',
  'historicalPattern': 'how this situation has played out in their past trades',
  'breathingRoom': 'one sentence of grounding/perspective for the trader',
}
```



```
'shouldStopTradingToday': boolean,
'stopReason': 'string or null'
}"
```

EMERGENCY BEHAVIORAL INTERVENTIONS: If `psychState = "TILT"` or `shouldStopTradingToday = true`: - Full-screen modal appears: cannot be dismissed for 60 seconds - Shows: “Your AI Coach has detected a high-risk emotional state” - Displays historical pattern: “The last 3 times you were in this state, you lost \$X” - Two buttons: “I’ll take a 10-minute break” | “I understand the risk, continue” - If user clicks continue: logs the override, tracks the outcome against their history

8.3 — THE LIVE COPILOT PANEL (UI)

A dedicated full-width panel that activates when a position is open.

Layout: 3-column grid + 1 full-width header

```
COPILLOT HEADER - Position Summary
AAPL LONG | Entry: $192.40 | Current: $194.10 | P&L: +$170 (+0.88%)
Time in trade: 47 min | Stop Loss: $190.00 | Target: $197.00
[AI analyzing...] [ ] LIVE
```

TECHNICAL (GPT-o3)	INSTITUTIONAL (Claude + Flow)	BEHAVIORAL MONITOR (Claude Opus 4)
Bias: BULLISH Strength: 72/100	Dark Pool: ACCUM Options: BULLISH GEX: PINNING	State: CALM_DISCIPLINED Risk Score: 18/100
RSI(14): 58 VWAP: Above	Gamma Flip: \$191	On plan Within time avg
ADX: 31 (trend) BB: Not squeezed	Smart Money: ALIGNED with you	Approaching your historical exit zone
Support: \$192.80 Resistance: \$195	Net Prem: \$2.1M bull \$0.7M bear	Likely mistake: EUPHORIA_HOLD (32%)
[View full calc]	[View flow feed]	[View full profile]

SOCIAL (Grok 4)	FUNDAMENTAL (Perplexity +	DARK POOL FEED
--------------------	------------------------------	----------------

	Gemini)	12:34:02 - 250K shrs
X Sentiment:		\$194.20 BUY side est.
BULLISH	Analyst:	LARGE PRINT
Score: +0.62	BUY \$201 PT	"Institutional buying near this level"
Dominant mood:	P/E: 28.4x	
Mild euphoria	Sector avg: 24x	12:21:18 - 89K shrs
Contrarian:	(slight prem.)	\$193.80 UNKNOWN
Retail is very bullish - risk of consensus unwind	Next earnings: 43 days away	STANDARD
	No near-term catalyst risk	12:09:44 - 1.2M shrs
		\$192.40 BUY side est.
		WHALE PRINT
[View X feed]	[Full analysis]	[View full feed]

COPILLOT CONSENSUS - Synthesized by Claude Opus 4

6/6 perspectives analyzed | Last updated: 23 seconds ago

OVERALL SIGNAL: HOLD POSITION - FAVORABLE CONDITIONS

"Technical momentum is bullish and trending above VWAP. The 1.2M share whale dark pool print at your entry suggests institutional accumulation at this level. Smart money options flow is \$2.1M net bullish. Your behavioral state is disciplined. Primary risk: retail sentiment is euphoric - watch for consensus unwind if price stalls at \$195 resistance."

STOP LOSS: \$190.00 (technically valid below key support)
NEXT DECISION POINT: \$195.00 (resistance - reassess at this level)

[Force Re-Analysis Now] [Copy to Journal]

8.4 — MULTI-TIMEFRAME RESEARCH PANEL

Route: /research/:symbol

A dedicated deep research page that can be opened for ANY symbol at any time — whether or not a position is open. This is the “institution-grade pre-trade research” workspace. Think Bloomberg terminal for a single ticker.

6 Tabs:

Tab 1: Technical Deep Dive (GPT-o3) - Full indicator suite across 6 time-frames (1m, 5m, 15m, 1h, 4h, Daily) - Support/resistance levels with mathematical derivation shown - Volume profile chart (price levels with most historical volume) - Statistical edge calculator: “At this price level, historically this stock has moved X% in Y direction Z% of the time” - Mean reversion calculator: probability of return to mean from current deviation

Tab 2: Institutional Intelligence (Claude + FlowAlgo) - 30-day dark pool print history chart (overlay on price) - Options flow 30-day heatmap (bullish vs bearish premium by day) - Gamma exposure curve chart (GEX across all strike prices) - Unusual options activity log (last 30 days, sortable by premium) - Institutional ownership changes (latest 13F filing changes)

Tab 3: Fundamental Deep Dive (Gemini 2.5 Pro) - Full company overview, business model summary - Financial metrics: Revenue, EPS, P/E, P/S, debt ratios, free cash flow - Earnings history: EPS beat/miss last 8 quarters - Analyst ratings history (upgrades/downgrades chart) - Industry comparison: how does this company rank in its sector on key metrics - Recent SEC filings summary (Gemini reads and summarizes 10-Q/10-K)

Tab 4: News & Social Timeline (Perplexity + Grok 4) - Verified news timeline for this ticker (last 30 days, Perplexity) - X sentiment score chart (daily Grok sentiment over 30 days) - Correlation overlay: did news events predict price moves? (statistical) - Finfluencer activity log: times this ticker was pushed on social media

Tab 5: Backtesting Stats (DeepSeek V3) For a selected setup/strategy on this ticker: - Historical win rate at similar price levels - Average holding time for winning vs losing trades - Best time of day/week to trade this specific ticker historically - Volatility patterns by day of week - Earnings event price reaction history (last 8 quarters)

Tab 6: AI Research Consensus (Claude) Claude synthesizes all 5 tabs into one unified research brief: - Bull case (top 3 arguments for the long side, with evidence) - Bear case (top 3 arguments for the short side, with evidence) - Key risk events in the next 30 days - Institutional positioning summary - Retail risk assessment: specific ways retail traders get hurt on this ticker - Overall context score: 0-100 (how well-researched is this trade opportunity)

8.5 — REAL-TIME MATHEMATICAL MODELS ENGINE

Powered by: GPT-o3 — All calculations shown with formulas

Pre-Trade Calculator (enhanced):

INPUT: Symbol, direction, entry price, stop loss, account size

GPT-o3 calculates and SHOWS the math:

1. RISK CALCULATION

Risk amount = Account $\times$ Risk% = \$50,000 $\times$ 2% = \$1,000
Distance to stop = |Entry - Stop| = |\$192.40 - \$190.00| = \$2.40
Max shares = Risk $\div$ Distance = \$1,000 $\div$ \$2.40 = 416 shares

2. KELLY CRITERION

$f^* = (bp - q) / b$
where $b = \text{avg_win} / \text{avg_loss} = 1.8$, $p = \text{win_rate} = 0.54$, $q = 1 - p = 0.46$
 $f^* = (1.8 \times 0.54 - 0.46) / 1.8 = 0.28$ (28% of bankroll)
Half-Kelly = 14% $\rightarrow$ \$7,000 position $\rightarrow$ 36 shares (conservative Kelly)

3. EXPECTED VALUE

$EV = (P_{\text{win}} \times \text{avg_win}) - (P_{\text{loss}} \times \text{avg_loss})$
 $EV = (0.54 \times \$432) - (0.46 \times \$240) = \$233 - \$110 = +\$123$ per trade
EV/Risk ratio: \$123 / \$1,000 = 0.123 (positive edge confirmed)

4. POSITION SIZE COMPARISON TABLE:

Method	Shares	Position \$	% of Account
Risk-based	416	\$80,070	160%
Half-Kelly	36	\$6,926	13.9%
2% Fixed	416	\$80,070	160%
Recommended	36	\$6,926	13.9%

5. BREAK-EVEN ANALYSIS

Commission (est.): \$1.00
Spread cost (est.): \$0.02/share $\times$ 36 shares = \$0.72
Total cost: \$1.72
Break-even move: \$1.72 / 36 shares = \$0.048 (+0.025%)

6. RISK:REWARD TABLE

Target	R:R	Win Rate Needed	Expected Value
\$194.80	1.0	50%	\$0
\$196.00	1.5	40%	+\$52
\$197.40	2.0	33.3%	+\$84
\$200.00	3.17	24%	+\$143

RECOMMENDATION: Target \$197.40 (2R) - minimum acceptable R:R for this setup

8.6 — LIVE BEHAVIORAL MONITORING DASHBOARD

Route: /live-monitor

A dedicated behavioral intelligence screen showing the trader's psychological state in real time across all active positions.

Components:

Emotional State Gauge: - 0-100 scale from CALM → ANXIOUS → EMOTIONALLY_COMPROMISED → TILT - Updates every 60 seconds based on position behavior - History chart: emotional state over the last 4 hours of trading

Behavioral Warning Cards (each trigger individually tracked):

TIME BREACH: Position held 2.3x longer than your average for this setup
LOSS PROXIMITY: Current P&L (-1.8%) approaching your panic exit pattern (-2.1%)
PEAK TRADING FATIGUE: Trade #8 today. Your win rate drops to 31% after trade #5
BAD WINDOW: You are trading in your historically worst window (2 PM-4 PM)
REVENGE RISK: 2 consecutive losses. You revenge traded after 2 losses in 7/10 cases
JOURNAL STREAK: 15-day streak maintained - your win rate is 28% higher when journaling

Decision Fatigue Tracker: - Tracks number of trades today vs user's historical optimal (when performance drops) - "You perform best on trades 1-4. This is trade 7. Your win rate after trade 5 is 31%."

Time-of-Day Risk Indicator: - Gradient heatmap of user's own historical win rate by hour - Big red zone over their worst trading window - Pulsing alert if they are currently trading in their red zone

Pattern Recognition Alerts (Claude cross-references live behavior to journal history):

"3 weeks ago when {{symbol}} was at a similar level and X sentiment was euphoric, you entered and exited at a 2.1% loss 23 minutes later. Current setup: similar."

8.7 — CRYPTO-SPECIFIC ON-CHAIN INTELLIGENCE

Additional data layer for crypto positions only

Data Sources: Glassnode API + Nansen API

On-Chain Research Panel (shown when position is a crypto asset):

WHALE MOVEMENT TRACKER (Glassnode):

- Exchange inflows/outflows last 4 hours
(large exchange inflows = whales moving to sell = bearish pressure)
- Large transaction count (>\$100K) last hour vs 30-day average
- Supply on exchanges % (increasing = distribution, decreasing = accumulation)
- SOPR (Spent Output Profit Ratio) - are people selling at profit or loss?
- MVRV Z-Score - is the asset overvalued or undervalued vs realized value?

SMART MONEY TRACKING (Nansen):

- Top 50 "Smart Money" wallets: net buyers or net sellers in last 4 hours
- Whale alert: any wallet with >\$10M moved in/out of exchange in last hour
- New wallet activity: is retail FOMO increasing (new wallet surge = top signal historically)
- DEX vs CEX flow: institutional activity tends to route through CEX

FUNDING RATE (Perpetual futures):

- Current funding rate: positive = longs paying shorts (crowded long)
negative = shorts paying longs (crowded short)
- High positive funding = warning: crowded long trade at risk of long squeeze
- Extreme values: >0.1% per 8 hours = historically precedes sharp reversal

Claude synthesizes all on-chain data:

"Whale activity and smart money are [aligned/opposed] to your position.

Exchange inflows have [increased/decreased] suggesting [distribution/accumulation].

Funding rate at {{fundingRate}}% indicates a [crowded/uncrowded] trade."

NEW API ROUTES (Add to Existing)

Live Copilot

GET /api/copilot/activate/:tradeId	→ trigger all 6 parallel AI jobs
GET /api/copilot/status/:tradeId	→ poll job completion status
GET /api/copilot/results/:tradeId	→ get all 6 perspective results
POST /api/copilot/refresh/:tradeId	→ force re-run all jobs
GET /api/copilot/consensus/:tradeId	→ get Claude synthesis

Dark Pool & Options Flow

GET /api/flow/darkpool/:symbol	→ dark pool prints (last 4 hours)
GET /api/flow/options/:symbol	→ options flow (last 4 hours)
GET /api/flow/gamma/:symbol	→ gamma exposure data
GET /api/flow/sweeps	→ all sweeps across market (live)
GET /api/flow/whales	→ whale prints across all symbols (live)

Deep Research

GET /api/research/:symbol	→ trigger full multi-AI research
GET /api/research/:symbol/technical	→ GPT-o3 technical analysis
GET /api/research/:symbol/institutional	→ institutional flow synthesis
GET /api/research/:symbol/fundamental	→ Gemini fundamental deep dive
GET /api/research/:symbol/social	→ Grok social intelligence
GET /api/research/:symbol/backtest	→ DeepSeek historical pattern stats
GET /api/research/:symbol/consensus	→ Claude full synthesis

Math Engine

POST /api/math/position-size	→ full Kelly + risk calc with shown math
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POST /api/math/expected-value	→ EV calculation for a setup
POST /api/math/risk-reward-table	→ R:R table at multiple targets
POST /api/math/options-price	→ options pricing + Greeks
POST /api/math/monte-carlo	→ Monte Carlo simulation for trade outcomes
 # Behavioral Monitor	
GET /api/behavior/live-state	→ current psychological state
GET /api/behavior/warnings	→ active behavioral warnings
GET /api/behavior/decision-fatigue	→ fatigue level + trade count analysis
GET /api/behavior/time-risk	→ current hour vs historical performance
POST /api/behavior/override-logged	→ log when user overrides a tilt warning
 # On-Chain (Crypto)	
GET /api/onchain/:symbol/whales	→ Nansen whale activity
GET /api/onchain/:symbol/exchange-flow	→ Glassnode exchange inflows/outflows
GET /api/onchain/:symbol/funding-rate	→ perpetual futures funding rate
GET /api/onchain/:symbol/smart-money	→ Nansen smart money net flow

NEW BullMQ QUEUES

Queue: copilot-technical	→ GPT-o3 (fires on position open + every 60s)
Queue: copilot-institutional	→ Claude + FlowAlgo data (same timing)
Queue: copilot-social	→ Grok 4 (fires every 5 min)
Queue: copilot-fundamental	→ Perplexity + Gemini (fires on open, then on news)
Queue: copilot-behavioral	→ Claude Opus 4 (fires every 60s)
Queue: copilot-consensus	→ Claude Opus 4 (fires when all 5 complete)
Queue: research-full	→ All models in parallel (on-demand)
Queue: math-calculation	→ GPT-o3 (on-demand, synchronous preferred)
Queue: onchain-scan	→ Glassnode + Nansen (every 5 min for crypto positions)

NEW SOCKET.IO EVENTS

Server → Client:

'copilot-perspective-ready'	→ {tradeId, perspective: 'technical' 'institutional' ..., data}
'copilot-consensus-ready'	→ {tradeId, consensusData}
'copilot-refresh-complete'	→ {tradeId}
'darkpool-print'	→ {symbol, print} - real-time whale print alert
'options-sweep'	→ {symbol, sweep} - real-time sweep alert
'behavioral-warning'	→ {warningType, severity, message}
'tilt-detected'	→ {stateScore, interventionRequired}
'onchain-whale-alert'	→ {symbol, walletAction, size}

Client → Server:

'copilot-subscribe'	→ {tradeId} - start live monitoring this trade
'copilot-unsubscribe'	→ {tradeId}
'research-subscribe'	→ {symbol} - subscribe to research updates
'tilt-override'	→ {tradeId} - log that user overrode tilt warning

NEW ENVIRONMENT VARIABLES

Dark Pool & Options Flow

UNUSUAL_WHALES_API_KEY=""	<i># unusualwhales.com</i>
FLOWALGO_API_KEY=""	<i># flowalgo.com</i>
QUANTDATA_API_KEY=""	<i># quantdata.us (alternative to flowalgo)</i>

Gamma Exposure

SPOTGAMMA_API_KEY=""	<i># spotgamma.com</i>
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On-Chain Intelligence (Crypto)

GLASSNODE_API_KEY=""	<i># glassnode.com</i>
NANSEN_API_KEY=""	<i># nansen.ai</i>

Funding Rate Data

BYBIT_API_KEY=""	<i># already may be present - for funding rate</i>
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NEW FRONTEND PAGES & COMPONENTS

app/(dashboard)/	
copilot/[tradeId]/page.tsx	← Live Copilot (activates on open position)
research/[symbol]/page.tsx	← Full Symbol Deep Research (6 tabs)
flow/page.tsx	← Live Dark Pool + Options Flow Feed
behavior/page.tsx	← Live Behavioral Monitor Dashboard
math/page.tsx	← Mathematical Models Calculator
components/	
copilot/	
CopilotPanel.tsx	← Main 6-panel live display
TechnicalPerspective.tsx	
InstitutionalPerspective.tsx	
DarkPoolFeed.tsx	← Live streaming prints
SocialPerspective.tsx	
FundamentalPerspective.tsx	
BehavioralMonitor.tsx	
CopilotConsensus.tsx	← Bottom synthesis bar
TiltIntervention.tsx	← Emergency modal
research/	


```

TechnicalDeepDive.tsx
InstitutionalDeepDive.tsx
FundamentalDeepDive.tsx
NewsTimeline.tsx
BacktestStats.tsx
ResearchConsensus.tsx
math/
PositionSizeCalculator.tsx ← Shows full GPT-o3 math
ExpectedValueCalc.tsx
RiskRewardTable.tsx
MonteCarlo.tsx
OptionsAnalyzer.tsx
behavior/
EmotionalStateGauge.tsx
BehavioralWarningCards.tsx
DecisionFatigueTracker.tsx
TimeOfDayRisk.tsx
PatternRecognitionAlert.tsx

```

COMPLETE AI COUNCIL UPDATED (v5)

TRADEGUARD AI COUNCIL v5

Claude Opus 4	Journal analysis, behavioral monitoring (live), daily reports, risk synthesis, copilot consensus, dark pool annotation, psych interventions
Grok 4	Real-time X/Twitter sentiment (every 5 min), finfluencer trap detection, pump group alerts, live narrative monitoring during open positions
Gemini 2.5 Pro	Earnings call transcripts, SEC filing analysis, fundamental deep dives, macro research synthesis, pre-market macro briefing
GPT-o3	ALL mathematical models (Kelly, position sizing, EV, R:R tables, options pricing, Monte Carlo), live technical indicator calculations, tax math
Perplexity API	Verified news with source citations (real-time), fact-checked fundamental data, regulatory filings
DeepSeek V3	Overnight batch analytics, historical backtesting

stats, weekly pattern computation (cost efficient)

OpenAI Embeddings Journal semantic search, news similarity matching,
text-embed-3 pattern cross-referencing across trade history

Total: 7 AI models, each deployed for the specific task where it is objectively best in class. No model is used where another performs better.

FINAL ACCEPTANCE CRITERIA FOR MODULE 8

Complete when: 1. Opening a position instantly activates the Live Copilot with 6 live panels 2. All 6 perspectives load within 15 seconds via parallel async processing 3. Copilot auto-refreshes every 60 seconds while position is open 4. Copilot force-refreshes immediately on >1% price move 5. Dark pool prints stream in real time for user's open positions 6. Options sweeps appear in real time across the full market 7. Behavioral Monitor fires a warning when any psychological risk threshold is crossed 8. Tilt Intervention modal appears and cannot be dismissed for 60 seconds when TILT detected 9. `/research/:symbol` delivers 6-tab full research in under 30 seconds 10. GPT-o3 position size calculator shows all intermediate math, not just the answer 11. Crypto positions show on-chain whale activity and funding rate in the copilot 12. Every AI-generated output is labeled with which model produced it and why 13. All copilot outputs are copy-to-journal with one click for post-trade analysis

AGENT NOTES FOR THIS MODULE

- The CopilotPanel fires 6 BullMQ jobs simultaneously — NOT sequentially. Each panel renders independently as its job completes (streaming UX)
- DarkPoolFeed uses Socket.io for true real-time streaming, NOT polling
- TiltIntervention modal must use a 60-second countdown timer that cannot be skipped
- The BehavioralMonitor MUST cross-reference the user's actual historical journal data — it is personalized, not generic. Generic warnings are worthless.
- GPT-o3 math output must ALWAYS show the formula + intermediate steps — the trader must be able to verify every calculation
- Research page tabs load in parallel — not waiting for each other
- All perspective results cached in Redis for 60 seconds (avoid redundant AI calls on refresh)
- On crypto positions, the onchain module only activates if the asset class is CRYPTO

- The Copilot Consensus panel waits for ALL 6 perspectives before rendering ““